

Alessio Pani

Mathematical Engineer



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Profile

MSc student at Politecnico di Milano, specializing in **Statistical Learning**, **Machine Learning** and **Data Analysis**. I combine a rigorous engineering mindset with the resilience and accountability forged through **12 years of competitive Judo** and **lifeguard experience**. My goal is to translate complex datasets into **strategic business decisions**. Currently seeking a **thesis internship** to apply mathematical and statistical models, quantitative methods, and machine learning techniques to solve real-world business challenges, combining analytical rigor with pragmatism.

Education

MSc in Mathematical Engineering - Statistical Learning Mar 2025 – Present
Politecnico di Milano

Key courses: **Applied Statistics**, Machine Learning, **Data Mining**, Stochastic Dynamical Models.

BSc in Mathematical Engineering Sep 2021 – Feb 2025
Politecnico di Milano

MSc Projects

Stochastic Modeling - Ising Model Sep 2026
Analyzed the convergence limits of the CAVI (Coordinate Ascent Variational Inference) algorithm, consolidating expertise in **variational inference** and honing **independent research** and **analytical problem-solving** skills.

Bayesian Statistics - PPMx Clustering Model Sep 2025 – Feb 2026
Group project implementing a **Gibbs Sampler** for a Product Partition Model with covariates (**PPMx**) for Bayesian clustering. Contributed to implementation in **Python**.

Fixed Rank Kriging (FRK) - Estimation of Italian municipal GDP Mar 2025 – Jul 2025
Developed a spatial statistical model (**FRK**) for municipal-level GDP estimation in Italy.
Acted as **Team Leader**, coordinating activities and implementing models in **R** and **Python** to bridge data gaps in official ISTAT records.

Work Experience

Lifeguard (Beach and Swimming Pool) Aug 2021 – Sep 2021
Ensured public safety through vigilant monitoring and **proactive risk assessment**, delivering timely emergency interventions.
Demonstrated a strong sense of responsibility and **crisis management** skills under pressure.

Technical Skills

Programming Languages: Python (Pandas, NumPy, Scikit-learn, SciPy, Statsmodels, Matplotlib), R (Tidyverse, ggplot2), Matlab, C, C++, SQL.

Tools: Git, LaTeX, Excel.

Modelling and Methods: Bayesian Inference, Spatial Statistics, Time Series Analysis, Game Theory, Predictive Modelling, **Stochastic Processes**.

Extra-curricular Activities

Model United Nations (GEMUN) Nov 2019
Selected for an interscholastic **Model United Nations (MUN) simulation**; represented a member state as a delegate, negotiating international resolutions and honing **mediation** and **public speaking**.

Judo (Competitive Level) 2008 – 2020
12 years of competitive Judo. Forged discipline, resilience, and strategic decision-making under pressure.
Built teamwork and trust-building skills through consistent high-intensity training (6 sessions per week).

Languages

English: C1 (TOEIC 2025)

Italian: Native